Ten-Gate Threshold Specification

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Abstract.

This specification template defines the ten VVUQ gate thresholds that bound robot-patient interaction code, as numbered shall-statements with acceptance criteria.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Index terms

1 Scope

This specification defines the threshold each of the ten gates enforces before an interaction may be generated or executed. It is normative; the requirements use shall.

2 Definitions

Terms used in the requirements.

`rification fraction` the share of verification checks that pass; admissible value is 1.0

`alidation agreement` the agreement of a gate's model with its bound standard

`CoV bound` the maximum admissible coefficient of variation across a sweep

3 Requirements

Requirements are identified REQ-Gxx.

`REQ-G01` the verification fraction shall equal 1.0 for every interaction

`REQ-G02` each gate shall meet its validation-agreement floor

`REQ-G03` the three catastrophe gates shall hold agreement 1.00 with no exception

Figure placeholder (`Images/threshold-forest.png`). The float, caption, and `\label` mirror the VVUQ-02 figures; replace the box with `\includegraphics`.

Figure 1: Figure 1. Placeholder forest plot of the ten gate thresholds.

4 Acceptance thresholds

Gate	Agreement floor	CoV bound
vascular no-fly	1.00 (hard)	tightest
human collision	1.00 (hard)	tightest
dispersion	floor per standard	within bound

Table 1: Table 1. Gate acceptance thresholds (modeled on VVUQ-05 Table 1).

Rights and Permissions

Cite This Specification

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Data Availability

Supporting records are openly available in the `kevinkawchak/cancer-automated` and `kevinkawchak/physical-ai-oncology-trials` GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

[1] K. Kawchak, “VVUQ Processing Priority over Code Generation: Method and Pipeline.” `https://doi.org/10.5281/zenodo.20372501`, 2026. `https://doi.org/10.5281/zenodo.20372501`.

[2] K. Kawchak, “Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ.” `https://doi.org/10.5281/zenodo.20421754`, 2026. `https://doi.org/10.5281/zenodo.20421754`.

[3] K. Kawchak, "Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510)." `https://doi.org/10.5281/zenodo.20535429`, 2026. `https://doi.org/10.5281/zenodo.20535429`.

[4] K. Kawchak, "National Physical AI Oncology Trial Platform." `https://doi.org/10.5281/zenodo.19244918`, 2026. `https://doi.org/10.5281/zenodo.19244918`.

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